

CSCE 5218 — DEEP LEARNING

Homework: Perceptron Learning with Binary Cross-Entropy

Total: 20 points | Student edition

The lecture derived a learning rule for a single neuron using squared-error loss. In this homework, you will derive a learning rule using **binary cross-entropy**, a loss commonly used for binary classification.

Consider a single sigmoid perceptron. Set $x_{k0} = 1$ and treat its coefficient w_0 as an ordinary weight. For training instance k ,

$$in_k = \sum_{i=0}^n w_i x_{ki} = \mathbf{w}^T \mathbf{x}_k, \quad y_k = \sigma(in_k) = \frac{1}{1 + e^{-in_k}},$$

where $t_k \in \{0, 1\}$ is the target. Define binary cross-entropy loss for this instance as

$$L_k = -[t_k \ln(y_k) + (1 - t_k) \ln(1 - y_k)].$$

Show your derivation, not only the final formulas.

1. Loss derivative

(3 points)

Derive $\partial L_k / \partial y_k$. Simplify your answer to a single fraction.

2. Parameter gradients

(5 points)

Using the chain rule and

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)),$$

derive $\partial L_k / \partial w_j$ for $j = 0, \dots, n$. Show all local derivatives. Simplify the final expression as much as possible, and explain the role of the $j = 0$ case.

3. Gradient-descent update

(3 points)

Write the unified gradient-descent update rule for every w_j , including w_0 , using learning rate η . Explain in one sentence how the sign of $t_k - y_k$ affects the direction of an update when $x_{kj} > 0$.

4. One numerical update

(6 points)

Let

$$\mathbf{x}_k = (1, 1, 2), \quad \mathbf{w} = (0, 0.2, -0.1), \quad t_k = 1, \quad \eta = 0.1,$$

where the first entry of $\mathbf{x}_k$ is the constant input $x_{k0} = 1$. Perform one gradient-descent update using binary cross-entropy. Report nonlinear values to four decimal places.

1. Compute in_k , y_k , and L_k .
2. Compute the gradients for w_0 , w_1 , and w_2 .
3. Compute the updated parameters.
4. Recompute the prediction and loss after the update.

5. Comparison with squared error

(3 points)

For a sigmoid neuron, squared-error loss produces a weight gradient containing the factor $y_k(1 - y_k)$, whereas the binary-cross-entropy gradient derived above does not contain this factor after simplification.

Explain why the factor cancels for binary cross-entropy and describe one possible training consequence when the sigmoid output is close to 0 or 1.